7 FAULT LOCATION

7.1 Overview

The control system CMC4+ includes the visual display of messages regarding the control events using car and landing position indicators and optionally in the control cabinet, either by connecting any of the diagnosis tools explained in "Programming features" or a position indicator to the connector X87 where we have the LS or PIU signal.

These messages appear in two types of situations:

- Control failure (see section on "Code interpretation")
- Special operating situations such as inspection, emergency, STOP switch activation, etc.

The information displayed has a fault indicator symbol (*) and an identifying character from 0 to 9 or from A to Z, only viewable with the PIU protocol. The data display will blink while the failure or special event persists.

For example: failure 0, the following visual display will appear:

***0**

In addition to the code given by the position indicator in case of a failure situation, the landing registers will show the following status:

In universal control (PB)	The will remain ON
In selective control (1BC/2BC)	They will remain OFF in failure situations in which registered calls are deleted

The direction arrows will always be turned off, in universal and selective control.

Some of these failures lock the system, i.e. after correcting the failure, it is necessary to activate the RESET push-button of the control board (if lockout was not performed for its repair) for the start-up procedure to be executed and to check its proper operation. The failures caused by parameters of the group cannot be eliminated with a simple RESET; values should be restored using the programming tool available, e.g. LD2T, POME, etc. Other failures that lock the system need to be released using the Emergency or Inspection switch.

In addition to the codes, all the boards of the CMC4+ system have a fault indicator LED. This LED encodes through pulses the state of the communications with the control board, indicating whether the teach-in has been carried out properly, if there is a failure in the CAN bus drivers or if everything is working correctly. For a detailed description, check the relevant board manuals.

7.2 Code interpretation

Besides the aforementioned failure code, the control CMC4+ also includes a number of related failure codes, which can be detected using the maintenance or diagnosis operating panel (e.g. LD2T, POME …). The corresponding failure code is also indicated.

7.2.1 FAULT 0: DOOR RUNNING-LIMIT SWITCH FAULT AND TEMPORARY LATCHING (SYSTEM LOCKED)

The car will be under failure situation when any door sensor is activated for more than 10 seconds or when incorrect sequences of door control, overload and immediate stop elements (VVVF) are detected. When the defective sensor situation is corrected, the control will immediately leave the failure situation and the lift will resume service.

The sensors considered are the following:

- Door photo-control (photocell or light curtain).
- Door open push-button.
- Door limit switches.
- Overload.
- Immediate stop (INMSTOP).
- Next trip not allowed (NOSTART).
- Temperature sensors (machine room/cabinet).

7.2.2 FAULT 1: CONTACTOR FAILURE (SYSTEM LOCK)

- If, when starting up, any contactor is connected, this failure will be dumped (LEDs UD SC LC turned off in status prior to start-up). When this happens, the lift will enter a system-locking failure situation. Once the failure is corrected, RESET should be carried out to resume service.
- Upon contactor failure, a test is nonetheless run for 400ms. If the release time of any of the contactors takes longer (LEDs UD SC LC turned off in stopped status), the lift will lock due to this failure until the maintenance staff intervene.
- Inverter failure (VVVF). Inverter not Ready. Inmstop signal.
- Control board failure.
- Incorrect parameter programming, if the traction configuration is not correct, this may cause a contactor failure. See traction type parameter.

7.2.3 FAULT 2: PHASE SEQUENCE FAILURE, BOTH SLOW-DOWN LIMIT SWITCHES ACTIVATED (SYSTEM LOCK)

Causes:

- Phase sequence failure, check the corresponding LED status on the control board.
- Detected both slowdowns activated, check the corresponding LED status on the control board.

7.2.4 FAULT 3: INSPECTION, RECOVERY AND SR MODULE

Causes:

- Failure in the SR module or related elements, detection by the floor Area or Zone signal. The signal sequence is not correct.

- Lift under inspection or recovery, functional status or wiring failures in inspection operating panel, car board failure, in reduced pit landing door opening sequence test relay.
- Control or car board fault.

7.2.5 FAULT 4: DOOR SERIES OPENING WHEN TRAVELLING (NON-LOCKING)

- Opening of the presence and/or interlock series while the lift is travelling. Check relevant LED status.
- Control board fault.

7.2.6 FAULT 5: DOOR OR SERIES OPENING FAULT (NON-LOCKING)

- The series of contacts of landing or car doors or landing door interlock fail to open in the time set subsequent to a command for them to open; possible fault in the operator unit, failure in the opening/closing relays located inside the car. Possible communication failure between control board and door operator control board.
- Control board fault.

7.2.7 FAULT 6: VANE COUNT ERROR (NON-LOCKING)

This failure appears when the lift stops beyond the stop vane due to incorrect counting or slipping. In this case, fault 6 is dumped for 5 seconds and is then corrected.

Possible causes:

- Incorrect activation/deactivation of the slowdowns.
- Incorrect reading of the vanes (or magnets), due to:
  - o Incorrect slowdown distances.
  - o Cuts in vanes or inadequate overlapping between them. Check the magnet position over guide rail.
  - o Inductors not properly aligned with vane line.
  - o Accumulated dirt (only on optical sensors).
  - o Defective inductors.
  - o Interference caused by inductive elements, contactor types, motors, coils, reactances.
  - o Fault in the vane order programming.

In lifts with regulated traction, a smaller stop vane than required (incorrect speed control system adjustment), the car loses the level and the photo switch goes out of the vane in boarding and disembarking processes.

Notes:

- In general, you should check the correct alignment and distribution of VANEs (magnets) according to the corresponding diagram for the Placement of Positioning elements, with or without preopening.

- Sensor operation should be checked as follows: taking into account the position of the sensor in the car ceiling, its activation is carried out placing the magnet in front of the sensor. When it is in front of VANE (magnet), the corresponding LED will be activated at the control board.
- Check the supply voltage to the sensor (24 V DC between the connector terminals of the car board).
- Check that there is voltage in the sensor output connector of the car board (magnet in front of the sensor).
- At the control board, we can check the sensor signals using the FS LEDs (upper) and FI (lower). In case of installations with a single sensor in the positioning system, the signal FI is activated.

7.2.8 FAULT 7: DOOR CLOSING FAILURE (NON-LOCKING)

If there is door closing order and the presence and/or interlock order is not reactivated, several attempts are made. The car and landing doors will physically close or not according to the failure situation or its cause.

Check at the board UBA2 the corresponding LED status when closing the presence and/or interlock series.

7.2.9 FAULT 8: NO VANE DETECTION; CAR COMMUNICATION FAILURE; LIFT IS NOT MOVING FROM LANDING LEVEL; SENSOR FAILURE WHEN LEVELLING (SYSTEM LOCK)

- Communication failure between control board and car board due to a wiring fault or board fault.
- No signal activity from the position sensors (FI or FS) is detected; its correct functioning is time-controlled in all the functional phases. The cause of this fault may be no sensor activity due to sensor failure (it does not detect the output and/or input in VANE) or due to there being no car movement when start-up conditions have been confirmed (traction fault).
- Check sensor operation, checking car board status using LED_RUN, and checking photo-switch connector.
- Failure of the emergency equipment. There is some problem with the remote alarm system according to EN81-28.

7.2.10 FAULT 9: PROGRAM ERROR; SECOND SERIES OPENING; POWER FAILURE (SYSTEM LOCK)

The control detects the opening of safeties series; in the absence of this signal, the lift locks awaiting the signal.

In hydraulic traction lifts, if the series opening takes place on an upper floor, the lift will lock until the series recovers and a reset is prompted (the ESH signal will be activated).

If the fault takes place without series opening, it may be due to the control board and/or altered parameters located in the Eeprom. In both cases, it is not possible to recover the lift without replacing the board and/or correctly parameterising the Eeprom.

7.2.11 FAULT F: PERIPHERAL ELEMENT NOT CONNECTED OR NOT CORRECTLY INITIALISED

The control detects if a peripheral element such as MP1, DMC, UCO, Teleservicio (remote service) ... should be connected in the control and it is not, or has not been correctly initialised.

7.3 List of failures

To find the failures, the information in the failure stack is arranged as follows:

All the codes are registered in the failure stack identified according to an order number in the list of 50 failures that can be stored by the CMC4+ control, the level at which the failure takes place, the travel direction, (indicated with an arrow ↑↓) and if the situation is P (lift stopped) or in M (lift running).

Depending on the type of fault code the information for the level will have a meaning other than the usual meaning.

Next the date appears, followed by the decimal code assigned to the failure and a brief description. In the case of outdated diagnosis terminals, if the failure is not known by the latter, we will find the correspondence between the code displayed and the failure caused in the following table.

The extended failure codes can only be properly viewed by using the PIU protocol. These codes extend the numeration from *A to *Z. In LS protocol these new codes are shown as any of those included in the LS table, e.g. *A is shown as G1, *B as GF and so on, going through the entire LS table until *Y is 7 and *Z is 8.

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
00	00h	No	NORMAL	**No**	Normal operation
01	01h	No	NO_ABRE_FPAF	**No**	Front open door limit does not open
02	02h	No	NO_ABRE_FPAT	**No**	Rear open door limit does not open
03	03h	No	NO_CIERRA_FPAF	**No**	Front open door limit does not close
04	04h	No	NO_CIERRA_FPAT	**No**	Rear open door limit does not close
05	05h	No	NO_ABRE_FPCF	**No**	Front closed door limit does not open
06	06h	No	NO_ABRE_FPCT	**NO**	Rear closed door limit does not open
07	07h	No	NO_CIERRA_FPCF	**NO**	Front closed door limit does not close
08	08h	No	NO_CIERRA_FPCT	**No**	Rear closed door limit does not close
09	09h	No	AP_FRON_ACT	**No**	Front door open push-button activated in excess
10	0Ah	No	AP_TRAS_ACT	**No**	Rear door open push-button activated in excess
11	0Bh	No	CAP_FRONT_ACT	**No**	Front capacitive edge activated

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
12	0Ch	No	CAP_TRAS_ACT	**No**	Rear capacitive edge activated
13	0Dh	No	FOT_FRONT_ACT	**No**	Front photocell activated
14	0Eh	No	FOT_TRAS_ACT	**No**	Rear photocell activated
15	0Fh	No	BORDE_FRONT_ACT	**No**	Front safety edge activated
16	10h	No	BORDE_TRAS_ACT	**No**	Rear safety edge activated
17	11h	No	NO_ABRE_41	**No**	The interlock series does not open with open order
18	12h	No	NO_ABRE_40	**No**	The presence series does not open with open order
19	13h	No	FALLO_E2PROM	**No**	Failure detected in EPROM memory
20	14h	No	NO_HAY_40	**No**	Presence series signal is not detected
21	15h	No	NO_HAY_41	**No**	Interlock series signal is not detected
22	16h	No	FALLO_SR	**No**	Fault in SR module detected (Non-locking)
23	17h	No	PRIORIDAD_ CABINA	**No**	Preference connection car
24	18h	No	FUNCIÓN _BOMBEROS	**No**	Fire service switch activated
25	19h	No	FUERA_SERVICIO	**No**	Out of service switch activated
26	1Ah	No	EARTHQUAKE_ON	**No**	Seismic sensor activated
27	1Bh	No	FALLO_COM_POSICION	**No**	Failure in position communication
28	1Ch	No	APER_SSEG_500	**No**	Safeties opening < 500ms
29	1Dh	No	APER_PTARUN_200	**No**	Opening door in operation <200ms
30	1Eh	No	Free	**No**	
31	1Fh	No	Free	**No**	
32	20h	No	EXCESO_CARGA	**No**	Overload sensor activation detected
33	21h	No	EXCESO_CARGA_DIF	**No**	Differential load sensor activation detected
34	22h	No	FALLO_MPLEX	**No**	Communication failure of the multiplex lift group
35	23h	No	CORRIENTE_EMERGENCI	**No**	Rotational emergency power function activated
36	24h	No	EXCESO_ANS	**No**	Expected uneven start-up time has been exceeded
37	25h	No	ACT_SENSOR_FUEGO	**No**	Fire sensor activated
38	26h	No	FALLO_MX_FS	**No**	Multiplex weighting fault, front calls going up
39	27h	No	FALLO_MX_FB	**No**	Multiplex weighting fault, front calls going down
40	28h	No	FALLO_MX_TS	**No**	Multiplex weighting fault, rear calls going up
41	29h	No	FALLO_MX_TB	**No**	Multiplex weighting fault, rear calls going down

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
42	2Ah	No	FALLO_SLAVES	**No**	Communication failure in multiplex
43	2Bh	No	FALLO_PROTEC_SALIDAS	**No**	Fault due to short-circuit in a protected output of the UCM2. More information in POME level
44	2Ch	No	POWER_FAIL	**No**	Power failure
45	2Dh	No	FALLO_24V	**No**	24V failure
46	2Eh	No	STS_SHABBAT	**No**	Control in SHABBAT mode
47	2Fh	No	FALLO_EQUIPO_EMERG	***8**	Emergency equipment in car (EAR) failure signal
48	30h	YES	ERROR_CONTAJE	***6**	Vane counting fault (Shaft copying)
49	31h	YES	OVER_CONEX	***0**	Number of connections/time exceeded
50	32h	YES	NO_CIERRA_41	***7**	No signal 41 when attempting to close door
51	33h	YES	NO_CIERRA_40	***7**	No signal 40 when attempting to close door
52	34h	YES	OVERT_DOOR	***7**	Door time exceeded
53	35h	YES	NO_NEXT_START	***0**	No start-up order activated
54	36h	YES	AMBOS_CM_ACC	***2**	Both activated slowdowns detected
55	37h	YES	PMS_SIN_PANT	***8**	Stop travel when upwards activated without vane(1V)
56	38h	YES	PMB_SIN_PANT	***8**	Stop travel when downwards activated without vane(1V)
57	39h	YES	APER_SERIE_SEG	***9**	Safeties series opening
58	3Ah	YES	APER_S40_MARCHA	***4**	Presence series opening
59	3Bh	YES	APER_S41_MARCHA	***4**	Interlock series opening
60	3Ch	YES	INCONGRUENCIA	***9**	Incorrect variables detected
61	3Dh	YES	CMS_ERROR	***6**	Slowdown when upwards incorrect activation B323
62	3Eh	YES	CMB_ERROR	***6**	Slowdown when downwards incorrect activation B322
63	3Fh	YES	APER_SEG_CMS	***9**	Opening with slowdown when upwards B323
64	40h	YES	APER_SEG_CMB	***9**	Opening with slowdown when downwards B322
65	41h	YES	PROGRAM_ERROR	***9**	Program error detected
66	42h	YES	SITUAC_INSPEC	***3**	Inspection control connection
67	43h	YES	SITUAC_RAPPEL	***3**	Recovery control connection
68	44h	NO	SITUACIÓN_CORREC	**NO**	Lift in correction phase
69	45h	YES	DETECT_LOAD.100	***0**	Overload activated
70	46h	YES	NO_COMUNIC_CAB	***8**	Car communication failure
71	47h	YES	DETECT_SOBREC_DIF	***0**	Differential overload activated
72	48h	YES	NO_AREA_RENIV	***3**	No area signal

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
73	49h	YES	CAPF_ACT_30S	*0	Front capacitive edge activated for 30 seconds
74	4Ah	YES	ON_MODOTEST	*9	Board UCM2 test mode activated
75	4Bh	YES	NO_COPYE2P	*9	Failure in the EEPROM parameter copying
76	4Ch	YES	APER_PTA_V0	*3	Landing door opened w/o voltage in main controller; reset activating and deactivating the emergency switch on the control board
77	4Dh	NO	CORTO_UCC	NO	Short-circuit in Car board UCC2. It may also trip due to overconsumption
78	4Eh	YES	APER_PTA_V1	*3	Landing door opened with voltage in main controller
79	4Fh	FREE	FREE		
80	50h	YES	CAPT_ACT_30S	*0	Rear capacitive edge activated for 30 seconds
81	51h	YES	FOTF_ACT_30S	*0	Front photocell activated for 30 seconds
82	52h	YES	FOTT_ACT_30S	*0	Rear photocell activated for 30 seconds
83	53h	YES	BORF_ACT_30S	*0	Front safety edge activated for 30 seconds
84	54h	YES	BORT_ACT_30S	*0	Back safety edge activated for 30 seconds
85	55h	YES	NO_UCPS_TRASERO	*A	Teach-in failure, rear landing board not detected, level shown in POME
86	56h	YES	NO_UCPS_FRONTAL	*B	Teach-in failure, front landing board not detected, level shown in POME
87	57h	YES	MAS_UCPS_FRONTAL	*A	Teach-in failure, more landing boards than programmed are detected (front)
88	58h	YES	MAS_UCPS_TRASERO	*B	Teach-in failure, more landing boards than programmed are detected (rear)
89	59h	NO	FALLO_NO_REGISTRADO		Faults not assigned
90	5Ah	YES	LOST_UCC	*C	Car communication failure
91	5Bh	YES	NO_INIT_UCC_FRONT	*D	Front Car board initialisation failure
92	5Ch	YES	NO_INIT_UCC_TRAS	*E	Rear Car board initialisation failure
93	5Dh	YES	SOBRETEMP_CTOMAQ	*0	Machine room overheating fault
94	5Eh	YES	SOBRETEMP_PTC	*0	PTC overheating fault
95	5Fh	YES	NO_CAE_CPIK06	*1	Running contactor K06 does not switch off. (CPI-CAN)

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
96	60h	YES	NO_ENTRA_CPIK06	*1	Running contactor K06 does not switch on. (CPI-CAN)
97	61h	YES	NO_CAE_CPIK0612	*1	None of the running contactors K06, K061 or K062 switch off. (CPI-CAN)
98	62h	YES	NO_ENTRA_CPIK0612	*1	None of the running contactors K06, K1 or K2 switch on. (CPI-CAN)
99	63h	YES	NO_CAE_CPIK0F	*1	Brake contactor K0F does not switch off. (CPI-CAN)
100	64h	YES	NO_ENTRA_CPIK0F	*1	Brake contactor K0F does not switch on. (CPI-CAN)
101	65h	YES	FALLO_CANCPI	*1	CAN communication failure between control and CPI. (CPI-CAN)
102	66h	YES	FALLO_SECCPI	*1	CPI Starting, stopping or slowdown sequence failure. (CPI-CAN)
103	67h	YES	FALLO_HNDCHKM16R8	*1	Internal communication failure. (CPI-CAN)
104	68h	YES	NO_ENTRA_BRK1	*1	Brake switch surveillance, contact does not switch on BRK1 (CPI-CAN)
105	69h	YES	NO_CAE_BRK1	*1	Brake switch surveillance, contact does not switch off BRK1 (CPI-CAN)
106	6Ah	YES	NO_ENTRA_BRK2	*1	Brake switch surveillance, contact does not switch on BRK2 (CPI-CAN)
107	6Bh	YES	NO_CAE_BRK2	*1	Brake switch surveillance, contact does not switch off BRK2(CPI-CAN)
108	6Ch	YES	FALLO_INITCPI	*1	Initialisation failure CPI CAN
109	6Dh	NO	SHAFT_FUME_SENSOR_ON	NO	Smoke detector in shaft activated
110	6Eh	YES	FALLO_CMI_A3	*3	Fault of unintended movement control system
111	6Fh	YES	INCOHERENCIA_CONTAJE	*8	Fault end section bi-stable
112	70h	NO	INUNDACION_SENSOR_ON	no	Buoy flooding sensor activated
113	71h	YES	ERROR_PARAM_CMI_A3	*3	Error in parameter ControlMI_A3
114	72h	YES	NIVEL_BAJO_ACEITE	*0	It has been detected that a hydraulic elevator has low oil level
115	73h	NO	LUZ_CABINA_SENSOR_ON	NO	Car light fault detected
116	74h	YES	PARADA_TELESERVICIO	*0	Command to stop received from remote service

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
117	75h	YES	BLOQUEO_EVACUACION	**NO**	After an automatic fire service evacuation as per parameterisation 32CFResetFireService the lift locks until it is reset
118	76h	YES	BLOQUEO_A3_FUERA_SR	***3**	An uncontrolled exit from the area is detected
119… 125	74h.. 7Dh	NO	free	**NO**	
126	7Eh	YES	NO_TMAXNUDGE	***9**	Programmed nudging time not within limits
127	7Fh	YES	NO_PLT_CORTAS	***9**	Short floor area incorrectly defined
128	80h	YES	INMEDIATE_STOP	***0**	Immediate stop
129	81h	YES	NO_IMPULSOS_PANT	***8**	No VANE impulses received
130	82h	YES	EXCESO_PEQUEÑA	***8**	Programmed time in slow exceeded
131	83h	YES	NO_ENTRA_HCONT	***1**	Fast speed contactor does not switch on
132	84h	YES	NO_CAE_HCONT	***1**	Fast speed contactor does not switch off
133	85h	YES	NO_ENTRA_LCONT	***1**	Slow speed contactor does not switch on
134	86h	YES	NO_CAE_LCONT	***1**	Slow speed contactor does not switch off
135	87h	YES	NO_ENTRA_UCONT	***1**	Upwards trip contactor does not switch on
136	88h	YES	NO_CAE_UCONT	***1**	Upwards trip contactor does not switch off
137	89h	YES	NO_ENTRA_DCONT	***1**	Downwards trip contactor does not switch on
138	8Ah	YES	NO_CAE_DCONT	***1**	Downwards trip contactor does not switch off
139	8Bh	YES	NO_CAE_UDCONT	***1**	Downwards trip contactor does not switch off
140	8Ch	YES	NO_PROX_PAR	***8**	Next stop not defined
141	8Dh	YES	NO_MOTORCODE	***9**	Traction type not defined
142	8Eh	YES	INVERSION_FASES	***2**	Direction reversal detected
143	8Fh	YES	FALLO_GRAB_EEPROM	***9**	EEPROM writing failure
144	90h	YES	ERROR_COLC_PANT	***8**	VANE placement error
145	91h	YES	NO_SALE_PANT	***8**	Start-up order and no VANE exit
146	92h	YES	FALLO_RAM	***9**	Reading/Writing RAM failure
147	93h	YES	FALLO_PARAM_EPROM	***9**	EEPROM parameter failure
148	94h	YES	NO_MAXT_DOOR	***9**	Maximum out of range door time
149	95h	YES	NO_CONEX_HORA	***9**	Maximum number of connections/time
150	96h	YES	NO_RESERVA_CONEX	***9**	Maximum connection reserve

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
151	97h	YES	NO_MAXT_GRANDE	*9	Maximum time at fast speed
152	98h	YES	NO_MAXT_PEQUEÑA	*9	Maximum time at slow speed
153	99h	YES	NO_MAXT_RENIVEL	*9	Maximum time Re-levelling
154	9Ah	YES	NO_CAB_CALL_MAX	*9	Maximum number of calls per car
155	9Bh	YES	NO_RET_STOP	*9	Maximum delay to stop
156	9Ch	YES	NO_TIEMPO_ET	*9	Maximum switching star-delta time
157	9Dh	YES	PULS_STUCK	*9	Stuck push-button, shows level in POME
158	9Eh	YES	NO_RET_RBB	*9	Maximum time delay after electrocam de-energisation
159	9Fh	YES	NO_PREFER_SUB	*9	Preferential upper level number
160	A0h	YES	NO_PREFER_INF	*9	Preferential lower level number
161	A1h	YES	NO_LOBBY_1	*9	Main level 1 number
162	A2h	YES	NO_LOBBY_2	*9	Main level 2 number
163	A3h	YES	NO_PLANTA_BOMBEROS	*9	Fire service floor not defined
164	A4h	YES	NO_ENTRA_DRDCONT	*1	Delta contactor does not switch on
165	A5h	YES	NO_CAE_DRDCONT	*1	Delta contactor does not switch off
166	A6h	YES	FALLO_CPI	*1	CPI regulator failure (Ready)
167	A7h	NO	FALLO-PULSOS	*8	Photo switch position sensor pulse fault
168	A8h	NO	CODE_FOT_F1	*8	Photo switch position sensor pulse fault (additional info)
169	A9h	NO	CODE_FOT_F2	*8	Photo switch position sensor pulse fault (additional info)
170	AAh	NO	CODE_FOT_F3	*8	Photo switch position sensor pulse fault (additional info)
171	ABh	NO	START_SIN_DIRECCION	*8	Starting command without setting the direction of travel
172	ACh	YES	FALLO_EQUIPO_MP1_TKE	*3	A required MP1_TKE module does not work
173	ADh	YES	SITUAC_RESCATE	*3	DEA rescue device activated
174	AEh	YES	RESCATE_OK	*3	DEA rescue finished
175	AFh	YES	RESCATE_ERROR	*3	Error in the DEA rescue control
176	B0h	YES	NO_COTAINTERMEDIA	*9	General fault in intermediate levels
177	B1h	YES	NO_COTAINTERINF	*9	Failure parameter lower intermediate level
178	B2h	YES	NO_COTAINTERSUP	*9	Failure parameter upper intermediate level
179	B3h	YES	FALLOCANHUECO	*1	Shaft CAN bus failure
180	B4h	YES	FALLOCANGRUPO	*1	Group CAN bus failure
181	B5h	YES	PARAM_ENCODER	*9	Error in a parameter related to the positioning with encoder
182	B6h	YES	MEDIDA_ENCODER	*9	Shaft measurement not valid

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
183	B7h	YES	PULSOS_ENCODER	*9	Encoder pulse signal fails
184	B8h	YES	SINCRO_ENCODER	*9	Positioning does not match with shaft measurements
185	B9h	YES	LOGICA_ENCODER	*9	Encoder general fault
186	BAh	YES	FALLO_CM_SUPLEMENTARIO	*9	The extra CM is not consistent with B322 “CMB” and B323 “CMS”
187	BBh	NO	FALLO_DMC	*F	The DMC device is not connected or has not been initialised properly
188	BCh	NO	FALLO_OP	*F	The UCO or door operator CAN is not connected or has not been initialised properly
189	BDh	NO	free	NO	
190	BEh	YES	FALLO_SECUENCIA_VACONCAN	*1	When stopping the Inverter does not authorise brake activation
191	BFh	YES	FALLO_SCNA_VACONCAN	*1	Failure to detect when stopping the opening of the series which is normally open
192	C0h	YES	FALLO_ESP_VACONCAN	*1	When stopping the Inverter does not authorise the opening of the motor contactor
193	C1h	YES	FALLO_SCNC_BRAKE_VACONCAN	*1	Failure to close when stopping the series normally closed
194	C2h	YES	FALLO_SCNCNAO_VACON_CAN	*1	At start-up the series which are normally open and the series normally closed are not in their initial status
195	C3h	YES	FALLO_SCNCNA_VACON_CAN	*1	At start-up the series which are normally closed fail to open
196	C4h	YES	FALLO_SECUANCIA_ESP_VACONCAN	*1	At start-up the Inverter does not authorise the activation of the motor contactor
197	C5h	YES	FALLO_SECUANCIA_EBS_VACONCAN	*1	At start-up the Inverter does not authorise brake release
198	C6h	YES	FALLO_SCNA_BRAKE_EBS_VACONCAN	*1	At start-up the series which are normally open do not close or do not open the brake micro switches
199	C7h	YES	FALLO_CAE_RUN_VACONCAN	*1	While running the inverter run signal fails
200	C8h	YES	SITAUCION_TEST_DE_FRENO	*3	The control is in brake testing mode
201	C9h	YES	ERROR_TEST_DE_FRENO	*1	Brake testing has been unsuccessfully completed
202	CAh	YES	FALLO_INTENSIDAD_INVERTER	*1	Unexpected current detected in the inverter transformer
203	CBh	YES	AVERIA_INVERTER	*1	Fault reported by the inverter

Code number			Fault-related text	Display	Meaning
Dec.	Hexa.	Activation Output FAULT			
204.. 250	CCh.. FCh	NO	free	**NO**	
251	FB	NO	RST_PILA_AVERIAS	**NO**	Fault stack has been reset
252	FCh	NO	INICIALIZANDO	**NO**	Control initialising
253	FD h	YES	RESET	**NO**	Manual reset completed
254	FE h	YES	FALLO254	**NO**	Generic failure
255	FF h	--	Not used	**--**	--

Failures with codes 85 and 86 are saved to the failure stack, storing the floor where the failure was detected in the byte for car position storage.

Failures with code 43 are saved to the failure stack, but in this case the car position byte will show the output where the short-circuit was detected. Thus:

- **24 to 17:** S_AUX_4, OMI2, OUT_Z, VINS, rele_rev, rele_ls, S_AUX_3, LS.
- **16 to 9:** TEACH_T, V2, OMI1, OUT_Y, SANS, rele_rbr, rele_hs, S_AUX_2.
- **8 to 1:** AVERIA, TEACH_F, No_Usado, EnFRENOTEST, RELE_HF49, T3_UCM, rele_up, rele_dw.
- **0:** general failure of short-circuit detection circuit.

When a short-circuit occurs, it can be detected in more than one output apart from the one where it occurred.

Test the LEDs described in the table shown in the “Normal Running” section.

The STATUS LED of the UBA2 is set to reveal certain failure situations by flashing. If this LED flashes, check the relevant table in the Assembly Manual.

8 REFERENCE DOCUMENTS

Additional information can be found in the following reference manuals:

- Overall Control Manual (group of boards UBA2, UCM2, UCV, VTH3…)
- Inverter Terminal Board Manual UBV
- Car Terminal Board Manual UBC
- Car board Manual UCC2
- Car operating panel board Manual BMC
- Car operating panel board Manual IMC
- Door Operator Board Manual UCO
- Programmable Module Manual MP1